DFN Parameterization and Validation of a High-Energy NCMA | Gr-SiO_x 21700 Cell

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Introduction

High-energy 21700 cells pairing Ni-rich **NCMA** cathodes with **Gr-SiO_x** composite anodes are attractive for automotive packs, but the hysteresis in the anode potential and coupled two-phase behavior make physics-based parameterization challenging.

We present an **end-to-end, open-source PyBaMM workflow** that builds and validates a Doyle-Fuller-Newman (DFN) model of the commercial BAK CG-50 cell at beginning of life (BOL), from electrode thermodynamics to pack-level drive-cycle prediction.

21700	~5 Ah	NCMA | Gr-SiO_x	2 kWh
BAK CG-50 cell	Nominal capacity	Cathode | anode	Immersion-cooled pack

Composite Anodes

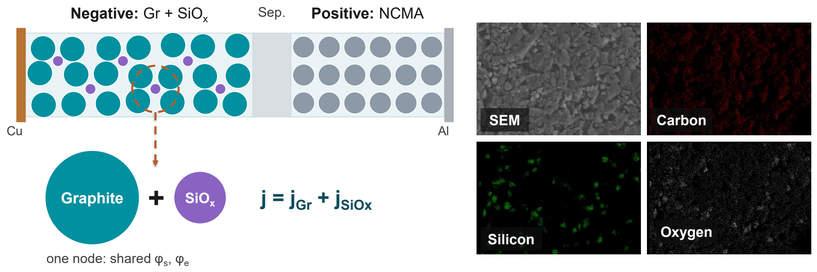
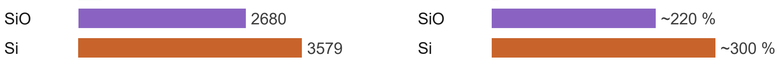


Fig. 1: (a) Two-phase composite anode in the DFN: graphite and SiO_x particles at each node share φ_s, φ_e; (b) SEM-EDS of the composite anode with C, Si and O maps

Specific capacity (mAh g^{-1})



Volume change on lithiation

Fig. 2: Theoretical specific capacity and lithiation volume change of graphite, SiO and Si [1, 2]

- Si/SiO_x raises capacity but swells comparatively more than graphite.
- Commercial anodes keep Si at ~10 wt% or less [1, 2].
- Si/SiO_x OCP differs substantially on lithiation vs. delithiation: **voltage hysteresis** [1, 2].
- **Two-phase model** with j = j_{Gr} + j_{SiOx}, available in PyBaMM [1, 2].

Cell Teardown and Characterization

- Commercial 21700 cells were torn down and the harvested cathode and anode rebuilt into coin cells with lithium metal as the reference electrode.
- **Half-cell OCPs** of the cathode and anode from C/30 cycling of these coin cells.
- **Electrolyte:** 1 M $LiPF_6$ in EC:EMC (3:7 v/v) with 2 wt% VC.
- **Solid-phase diffusivities** over lithiation and delithiation, and across temperature.
- **Exchange current densities** from EIS at different SOC and temperature.
- **SEM-EDS** (Fig. 1b): SiO_x weight fraction ≈ 4.5–5.7 % in the composite anode; ≈ 87 % nickel in the cathode.

PyBaMM DFN Setup

Two-phase anode · Gr + SiO_x	**SiO_x hysteresis** · one-state (Axen)
Geometry · cylindrical	**Contact resistance** · included
Kinetics · symmetric Butler-Volmer	**Diffusivity** · SOC and temperature dependent
Thermal · lumped	**Aging** · SEI, loss of active material, plating

OCP Deconvolution

In the multi-site, multi-reaction (MSMR) framework, each host is a set of galleries j (Li insertion reactions) with its own equilibrium potential [3]:

$$\mathrm{Li}^+ + \mathrm{e}^- + \mathrm{H}_j \rightleftharpoons (\mathrm{Li}-\mathrm{H})_j$$

$$U_j = U_j^0 + \frac{\omega_j}{f}\ln\left(\frac{X_j - x_j}{x_j}\right)$$

$$x_j = \frac{X_j}{1+\exp\left[f\left(U_{\mathrm{ocp}} - U_j^0\right)/\omega_j\right]}$$

- X_j: fraction of all host sites in gallery j
- x_j: fraction of gallery j's sites that are filled
- U_j^0, ω_j: gallery potential and non-ideality
- f = F/RT

Step 1 · Graphite (7 galleries)

A 7-gallery MSMR model is fitted to the measured graphite OCP, then held fixed.

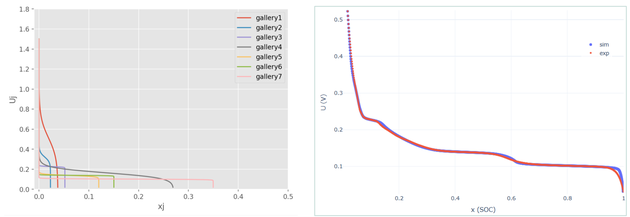


Fig. 3: Graphite MSMR fit: (a) gallery potentials vs. filled fraction x_j; (b) measured (red) vs. fitted (blue) graphite OCP

Step 2 · Gr–SiO_x blend (4 SiO_x galleries)

The blend shares its capacity through the SiO_x capacity fraction γ_q [3]:

$$(1-\gamma_q)\sum_{k=1}^{K_{\mathrm{Gr}}} X_{k,\mathrm{Gr}} + \gamma_q \sum_{k=1}^{K_{\mathrm{SiO}}} X_{k,\mathrm{SiO}} = 1$$

$$\gamma_q = \frac{w_{\mathrm{SiO}}\, q_{0,\mathrm{SiO}}}{w_{\mathrm{SiO}}\, q_{0,\mathrm{SiO}} + (1-w_{\mathrm{SiO}})\, q_{0,\mathrm{Gr}}}$$

Silicon Oxide Capacity Fraction

$w_{\mathrm{SiO}} \approx 5\,\%$ $\quad q_{0,\mathrm{SiO}} = 1750\ \mathrm{mAh\,g^{-1}}$

$q_{0,\mathrm{Gr}} = 340\ \mathrm{mAh\,g^{-1}}$

$$\gamma_{q,0} = \frac{0.05\times 1750}{0.05\times 1750 + 0.95\times 340} \approx 0.21$$

With graphite fixed, the SiO_x galleries and γ_q are optimized until the blend matches the measured composite-anode OCP, giving **independent Gr and SiO_x OCPs**.

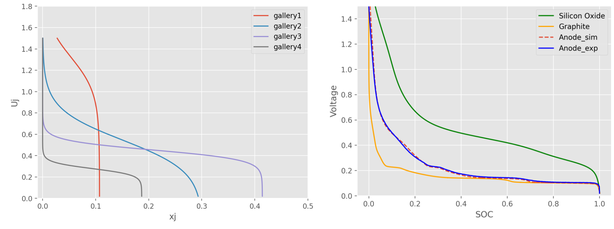
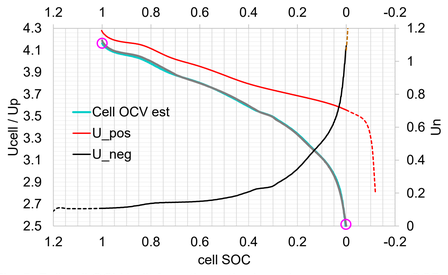


Fig. 4: (a) SiO_x gallery potentials; (b) composite-anode OCP rebuilt from Gr and SiO_x, simulated vs. measured

Full-Cell OCV Balancing



$$U_{\mathrm{cell}}(\mathrm{SOC}) = U_p(y) - U_n(x)$$

$$x = x_f + \mathrm{SOC}\,(x_i - x_f)$$

$$y = y_f - \mathrm{SOC}\,(y_f - y_i)$$

$$Q_{\mathrm{neg}}^T = \frac{Q_{\mathrm{cell}}^T}{x_i - x_f} \qquad Q_{\mathrm{pos}}^T = \frac{Q_{\mathrm{cell}}^T}{y_f - y_i}$$

Fig. 5: Full-cell OCV built from the cathode and composite-anode OCPs within the fitted stoichiometry windows (x, y: anode and cathode lithiation; i, f: 100 % and 0 % SOC).

Validation and Results

11 mV	12.07 mV	9.12 mV
Graphite MSMR fit RMSE	Composite-anode OCP RMSE	Full-cell OCV RMSE

(i) Full-cell quasi-equilibrium OCV

Low-current (C/50) charge and discharge in PyBaMM reproduce the measured full-cell OCV.

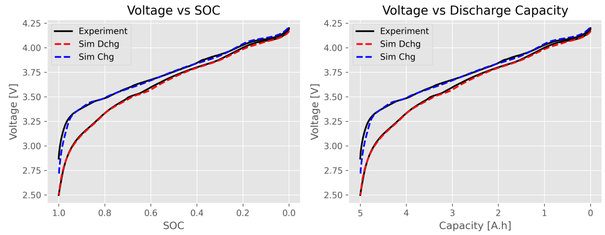


Fig. 6: Full-cell C/50 OCV vs. SOC and capacity, DFN vs. experiment

(ii) Rate capability

CC discharge and CC-CV charge at 0.5C, 1C and 1.5C, compared with capacity tests.

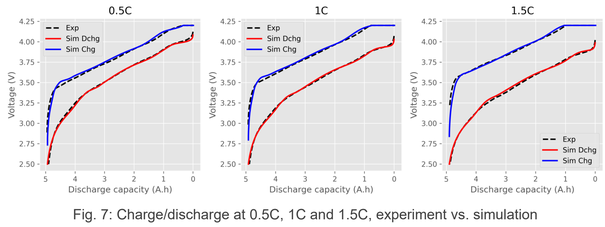


Fig. 7: Charge/discharge at 0.5C, 1C and 1.5C, experiment vs. simulation

(iii) Pack drive cycle

BMS-measured current from the immersion-cooled 2 kWh module is applied as a current interpolant; voltage is compared point-by-point.

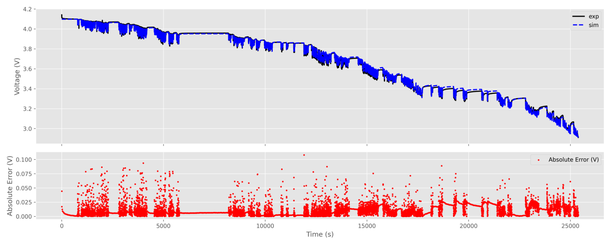


Fig. 8: Scaled-down pack drive-cycle voltage and absolute error

Conclusions

- OCP deconvolution with an SEM-EDS-based capacity fraction gives independent Gr and SiO_x open-circuit potentials (OCPs).
- PyBaMM low-current calibration reproduces the full-cell OCV (9.12 mV RMSE).
- The two-phase DFN tracks the C-rate tests to an average 15 mV RMSE and a real pack drive cycle to 14.8 mV RMSE.

References

[1] W. Ai, N. Kirkaldy, Y. Jiang, G. Offer, H. Wang, B. Wu, "A composite electrode model for lithium-ion batteries with silicon/graphite negative electrodes," *J. Power Sources* 527 (2022) 231142.

[2] M. P. Bonkile, Y. Jiang, N. Kirkaldy, V. Sulzer, R. Timms, H. Wang, G. Offer, B. Wu, "Coupled electrochemical-thermal-mechanical stress modelling in composite silicon/graphite lithium-ion battery electrodes," *J. Energy Storage* 73 (2023) 108609.

[3] M. Tu, T.-S. Dao, M. W. Verbrugge, B. Koch, "Mathematical model for a lithium-ion battery with a SiO/graphite blended electrode based on a reduced order model derived using perturbation theory," *J. Electrochem. Soc.* 171 (2024) 050539.